Omar Adel

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Education & Training

New Mansoura University Egypt
Bachelor of Science in Computer Science *2023 – 2027*

- **Relevant Coursework:** Computer Networks, Machine Learning, Artificial Neural Networks, Cloud Computing, Data Warehousing, Digital Logic Design, Deep Learning, Databases.
- **Professional Training:** Flutter & Dart Development (Route IT Academy) | Mastering Clean Architecture & SOLID Principles (Usama Elgendy).

Technical Skills

Languages: Dart, C#, Python, C++, SQL, Bash / Shell Scripting, HTML/CSS
Frameworks & Libraries: Flutter, Firebase, .NET Core, Flask, Streamlit, BLoC / HydratedBLOC
Databases & Storage: SQL Server, SQLite, Cloud Firestore, Data Warehousing Fundamentals
Developer Tools & Platforms: Git, GitHub Actions (CI/CD), Docker, Linux (Fedora CLI), Shorebird, Figma, Plane
Architecture & Workflows: RESTful APIs, Cross-Platform Architecture, Local Speech Recognition, Terminal Utilities (fzf, Multiplexers)
AI Developer Tooling & Agentic Workflows: Claude Code, OpenCode, Gemini CLI, Codex, VsCode Agent, Antigravity IDE)

Key Projects

Daftari | *Flutter, C# .NET, Turso, Cloudflare Workers, Payment Gateway, Python, PostHog, GitHub Actions* 2026

- Architected an offline-first Windows POS desktop application pairing a Flutter UI with C# .NET 8 background services for low-latency hardware contol and integration and integrated payment terminal SDKs and payment gateways to process card/digital transactions with automated invoice reconciliation and instant receipt printing.
- Engineered a multi-tenant edge database system on Turso (libSQL) using Cloudflare Workers as a serverless gateway to manage request routing, authentication validation, and secure communication with edge databases, implementing automated sharding under 1GB free-tier constraints.
- Built a Python decision engine and async migration executor for zero-downtime tenant extraction, dynamic client reconfigurations, and table-level sharding (separating receipts, archives, and sessions).
- Enforced zero-data-loss guarantees using transactional rollbacks and checksum verifications while embedding PostHog telemetry and automated CI/CD via GitHub Actions and Inno Setup.

BudgetWise | *Flutter, Dart, BLoC / HydratedBLOC, SQLite, Offline STT, SMS Scraping* 2026 – Current

- Architected a multi-asset finance tracking app with BLoC/HydratedBLOC, supporting custom account types (Credit, Debit, Meeza, Gold commodity tracking, Cash) and real-time net-worth calculation.
- Engineered an inter-account transaction ledger to track InstaPay transfers, peer-to-peer payments, and automated cross-account debt deductions.
- Built a dynamic analytics engine featuring visual expense statistics, recurring subscription tracking, savings goals, and CSV data export utilities.
- Integrated automated SMS transaction scraping and local offline speech-to-text models for hands-free voice expense logging.

Voice Assistant | *Python, Flutter, Dart, WebSockets, STT/TTS, REST API, LLM Integration* 2026

- Architected a full-stack voice platform pairing a Flutter mobile client with a Python speech processing backend.
- Engineered a speech pipeline integrating speech-to-text (STT) transcription, intent/command parsing, and text-to-speech (TTS) audio synthesis.
- Implemented real-time audio chunking and WebSocket/REST communication protocols to reduce end-to-end latency during active voice sessions.
- Built a dynamic UI featuring interactive audio visualization, session history management, and offline voice action fallback routines.

Obsidian Nexus | *Node.js/TypeScript, Gemini CLI, Obsidian API, Markdown, Git* 2026

- Developed an open-source Gemini CLI extension to integrate AI workflows with local Obsidian knowledge bases.
- Engineered a graph-aware markdown parser to resolve bidirectional [[Wikilinks]], YAML frontmatter, and nested tags for contextual prompt injection.
- Implemented custom slash commands enabling bidirectional vault operations, automated daily note entries, and smart frontmatter metadata generation.
- Optimized local file system indexing with AST parsing to query multi-thousand note vaults with minimal token overhead and low latency.